R 5.0

28 **B** Originally, $eV_s = hf - \Phi$, $V_s = (hf - \Phi)/e$
Intensity L = power/area = nhf/At
Frequency $2f$ implies energy of photon is doubled. Same metal surface used implies work function, Φ remains the same. Thus maximum KE of electron or its PE gained, $eV'_s = (2hf - \Phi)$.
Thus new stopping potential $V'_s = (2hf - \Phi)/e > 2(hf - \Phi)/e$ or $2V_s$

Intensity remains the same, but since frequency of light is now $2f$, number of photons incident on the metal surface is halved.
Thus the number of photoelectrons emitted per unit time will be less; maximum photocurrent will be less than I.